GalaxyGAN: Conditional Generation of Galaxy Morphologies with WGAN-GP

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GitHub URL

`https://github.com/ndhanrajani/GalaxyGAN`

Project Overview

This project explores class-conditional image generation using a Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP) model trained on data comprising images of galaxies. Specifically, this project aims to design a class-conditional generator model capable of producing galaxy images with the desired morphology class labels while achieving consistent model training on Northwestern Quest infrastructure.

The project uses the **Galaxy10 DECaLS** dataset from AstroNN. The dataset comprises $\tilde{1}$7,700 images of galaxies belonging to 10 different morphology classes: mergers, smooth ellipticals, spirals, and edge-on galaxies. It is a great dataset to use for this project due to the distinct appearance of each class.

A generator architecture based on DCGAN was chosen; class conditioning is performed using embeddings of the morphology class label. The inputs to the generator comprise:

- 128-dimensional latent vector
- Learned embeddings of the galaxy morphology class label

Inputs to the discriminator (critic) besides those above include embeddings of galaxy morphology class label distributed throughout the image grid and concatenated with the input image tensor.

The output of the generator is a 69×69 RGB image. To accommodate the non-square target size, the generator produces an output of size 64×64 which is then resized to the necessary dimensions.

All models and training processes were implemented in PyTorch using Slurm batch jobs on Northwestern Quest GPU nodes.

Extra Criteria Pursued

1. Conditional Generation by Morphology (Completed)

Conditioning on the morphology label was done using embedding of the class label in the generator and discriminator. Sample grids were created for all 10 morphological classes with outputs varying based on the conditioning label. The generated images contained galaxy-like structures, such as bright cores, elongated edge-on structures, diffuse merger-like objects, and smooth elliptical.

2. Interactive GUI via Gradio (Completed)

A Gradio-based interface was built for interactive use of the trained GAN allowing users to select a galaxy morphology class, provide a seed value, and generate a corresponding galaxy image from a specific GAN checkpoint.

3. Training Evaluation and Analysis (Partially Completed)

Qualitative analysis of the training procedure using sample grids and model checkpoints saved throughout the process was provided. Model outputs of curated samples were analyzed at epochs

$$1,\ 50,\ 100,\ 150,\ 200,\ 300,\ 1200$$

The following observations were made:

- Structure learning occurred in earlier epochs.
- Peak performance was recorded in intermediate epochs.
- Degradation began occurring during later stages.

Although a more quantitative approach to evaluation involving FID and Inception Score was planned initially, it could not be completed due to lack of time.

Difficulties Faced and Solutions

1. CUDA and PyTorch Compatibility Problems

One of the main issues with running the code on Northwestern Quest was compatibility of PyTorch CUDA. The initial code was executed using a wrong PyTorch CUDA package and exhibited issues like CUDA initialization errors, silent fallback to CPU mode, and very slow training. The problem was solved by reinstalling PyTorch with the correct CUDA version and implementing a fail-fast approach to stop jobs that run into CUDA problems.

2. Training Speed

At the beginning, the training was extremely slow with multiple hours needed per epoch. Some of the reasons for that were:

- Multiple read-throughs of data stored in common HDF5 file.
- Excessively expensive resizing operation performed by the data loader.
- Data loader parameter num_workers = 0.

Several approaches were taken to mitigate these issues:

- Pre-loading images into memory.
- Copying dataset onto a local node disk drive.
- Applying multiprocessing optimizations to the DataLoader.
- Enabling persistent workers.

3. GAN Instability and Collapse

The first stage of training revealed classic signs of GAN training instability, such as checkerboard effect, noise, and low diversity in the output. Peak performance was reached around epochs 100–150 after which the output started to degrade. This issue was resolved by adjusting hyperparameters, namely lowering learning rate and reducing the number of critic updates per iteration (n_critic = 3). The process was monitored using qualitative evaluation of the output at each epoch. This helped determine an early stopping criteria.

Results

The trained generator produces images resembling galaxy structures without exhibiting any constant noise patterns. Some characteristics of images produced include:

- Bright central core
- Diffuse light distribution
- Elongated, edge-on galaxy shape
- Characteristics associated with mergers.

The best results are attained with epochs 100–150 using:

$$\texttt{n\_critic = 3}, \quad \texttt{learning rate = } 1 \times 10^{-4}$$

Although the images are low-resolution and somewhat noisy, the project successfully demonstrates conditional image generation and meaningful structure learning on galaxy morphology data.

Future Work

Some directions to pursue in the future include:

- Substitution of transposed convolutions for up-sampling + convolution operations to alleviate the checkerboard effect.
- Using more quantitative criteria, e.g. FID and Inception Score.
- Performing analysis of latent spaces using techniques like interpolation, PCA, t-SNE.